

1. **Ans:** $5x + 7 + \frac{21}{x-2}$.
Work: $5x^2 - 3x + 7 = (5x + 7)(x - 2) + 21$.
2. **Ans:** $2x^2 + 5x - 10 + \frac{16}{x+1}$.
3. **Ans:** $2x^2 + 0x + \frac{5x-1}{3x-2}$ i.e., $2x^2 + \frac{5x-1}{3x-2}$.
Check: $(3x-2)(2x^2) = 6x^3 - 4x^2$; remainder $5x - 1$.
4. **Ans:** $\frac{x^3-1}{x-1} = x^2 + x + 1$ (remainder 0). End behavior is like the polynomial $x^2 + x + 1$ (no HA; grows like x^2).
5. **Ans:** $x^2 + 2x + \frac{-1}{x}$. End behavior grows like $x^2 + 2x$.
6. **Ans:** Zeros from numerator: $x = 1, 4$, provided denominator $\neq 0$. Denominator zeros are $x = \pm 3$, so both 1 and 4 are valid x -intercepts.
7. **Ans:** Simplify $g = \frac{(x+5)(x-2)^2}{(x-2)(x+1)} = \frac{(x+5)(x-2)}{x+1}$ with a hole at $x = 2$.
 x -intercepts: $x = -5$ (simple $\Rightarrow$ cross), and $x = 2$ is *not* an intercept (hole removed).
8. **Ans:** $\frac{x^2-4x+3}{x-3} = \frac{(x-1)(x-3)}{x-3} = x-1$ with a hole at $x = 3$.
 So the simplified function is $x-1$. x -intercept at $x = 1$ (cross).
9. **Ans:** *True*. Cancelling creates a hole, so that x -value cannot be an x -intercept of the simplified function.
10. **Ans:** $\frac{x^2-1}{(x-1)(x-4)} = \frac{(x-1)(x+1)}{(x-1)(x-4)} = \frac{x+1}{x-4}$ with a hole at $x = 1$.
 VA at $x = 4$ only.
11. **Ans:** $\frac{(x-3)^2}{(x-3)(x+2)} = \frac{x-3}{x+2}$ with a hole at $x = 3$.
 VA at $x = -2$.
12. **Ans:** $\frac{x^2-16}{x+4} = \frac{(x-4)(x+4)}{x+4} = x-4$ with a hole at $x = -4$.
No vertical asymptotes.
13. **Ans:** $\frac{(x+1)(x-2)}{x(x-2)^2} = \frac{x+1}{x(x-2)}$ with a hole at $x = 2$.
 VAs at $x = 0$ and $x = 2$? After cancellation, the factor $(x-2)$ remains once in the denominator, so VA at $x = 2$ *and* VA at $x = 0$.
14. **Ans:** Hole at $x = -2$. Simplified $F(x) = \frac{x-6}{x-1}$.

15. **Ans:** $\frac{x^2 - 9}{x^2 - 6x + 9} = \frac{(x - 3)(x + 3)}{(x - 3)^2} = \frac{x + 3}{x - 3}$ with a hole at $x = 3$. No cancellation remains at $x = -3$, so no hole there.
16. **Ans:** $\frac{(x - 4)^2}{(x - 4)} = x - 4$ with a hole at $x = 4$. There is *no* VA at $x = 4$ (factor cancels).
17. **Ans:** Degrees equal $\Rightarrow$ HA $y = \frac{4}{2} = 2$.
18. **Ans:** $\deg N = 3$, $\deg D = 2 \Rightarrow$ slant (linear) asymptote. Long division: $\frac{3x^3 - x + 1}{x^2 - 1} = 3x + \frac{3x + 1}{x^2 - 1}$.
Slant $y = 3x$.
19. **Ans:** $\deg N < \deg D \Rightarrow$ HA $y = 0$.
20. **Ans:** Long division: $\frac{x^4 - 3x^2 + 1}{x^2 - 2} = x^2 - 1 + \frac{-x^2 + 3}{x^2 - 2}$.
End behavior like $x^2 - 1$ (parabolic growth).